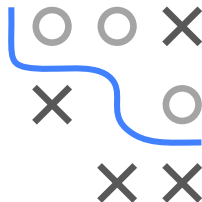


MULTI-TARGET PREDICTION: MOTIVATION

- Conventional supervised learning: Label space $\mathcal{Y}$ is 1-D.
- Multi-target prediction (MTP): multiple targets of mixed types (binary, nominal, ordinal, real-valued).
- Learn one model per target independently? $\rightsquigarrow$ Targets can be *statistically dependent*.
- Multi-label Emotions Dataset: 4 emotions of a music piece. Multiple emotions may be attributed to a single piece. Mutual information of the labels are:

	Calm	Quiet	Sad	Angry
Calm	1.000	0.073	0.018	0.290
Quiet	0.073	1.000	0.241	0.164
Sad	0.018	0.241	1.000	0.067
Angry	0.290	0.164	0.067	1.000

- It might be better to tackle targets *simultaneously*.

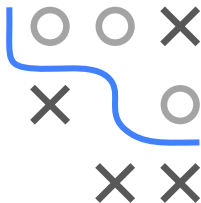


MULTI-TARGET PREDICTION: CHARACTERISTICS

Characterized by instances $\mathbf{x} \in \mathcal{X}$ and targets $m \in \{1, 2, \dots, l\}$ with following properties:

- A training set $\mathcal{D} = \{(\mathbf{x}^{(i)}, \mathbf{y}^{(i)})\}_{i=1}^n$, where $\mathbf{y}^{(i)} = (y_1^{(i)}, \dots, y_l^{(i)})$, with $y_m^{(i)} \in \mathcal{Y}_m$ is label for target m .
- n instances and l targets $\rightsquigarrow$ Labels $y_m^{(i)}$ can be arranged in an $n \times l$ matrix $\mathbf{Y}$. Note $\mathbf{Y}$ may have missing values.
- Target spaces $\mathcal{Y}_m$ can be nominal, ordinal or real-valued.
- Goal: predict scores for any pair $(\mathbf{x}, m) \in \mathcal{X} \times \{1, 2, \dots, l\}$.

In conventional MTP setting: no available side information for targets.



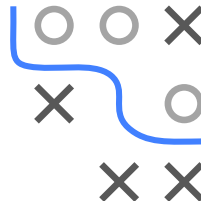
MULTIVARIATE REGRESSION

Target space $\mathcal{Y}_m = \mathbb{R} \forall m \in \{1, 2, \dots, l\}$.

		Mol1	Mol2	Mol3	Mol4	Mol5	Mol6
01101		1,3	0,2	1,4	1,7	3,5	1,3
00111		2	1,7	1,5	7,5	8,2	7,6
01110		0,2	0	0,3	0,4	1,2	2,2
10001		3,1	1,1	1,3	1,1	1,7	5,2
01011		4,7	2,1	2,5	1,5	2,3	8,5
11110		?	?	?	?	?	?

Waegeman et al. (2019), Multi-target prediction: A unifying view on problems and methods ([URL](#)).

Example: Predict binding strength between proteins (rows) and molecules (columns).



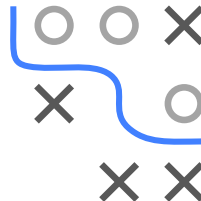
MULTI-LABEL CLASSIFICATION

Target space $\mathcal{Y}_m = \{0, 1\} \forall m \in \{1, 2, \dots, l\}$

		Tennis	Football	Biking	Movies	TV	Belgium
01101	Text1	0	1	0	0	1	1
00111	Text2	1	0	0	0	0	1
01110	Text3	0	0	0	1	1	0
10001	Text4	0	0	1	0	1	0
01011	Text5	1	0	0	1	0	0
11110	Text6	?	?	?	?	?	?

Waegeman et al. (2019), Multi-target prediction: A unifying view on problems and methods ([URL](#)).

Example: Assign documents (rows) to category tags (columns).



LABEL RANKING

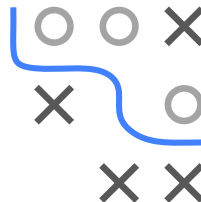
In *label ranking*, each instance is associated with a ranking of targets.

$\mathcal{Y}_m = \{1, \dots, l\} \forall m$, and labels (i.e., ranks) $y_m^{(i)} \neq y_k^{(i)} \forall m \neq k$.

		Tennis	Football	Biking	Skating	Running	Walking
01101	User 1	2	1	4	3	5	6
00111	User 2	1	4	3	5	6	2
01110	User 3	4	5	1	2	3	6
10001	User 4	4	3	2	6	1	5
01011	User t5	1	3	5	2	6	4
11110	User 6	?	?	?	?	?	?

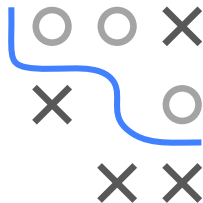
Waegeman et al. (2019), Multi-target prediction: A unifying view on problems and methods ([URL](#)).

Example: Predict for users (rows) their preferences over specific activities (columns).



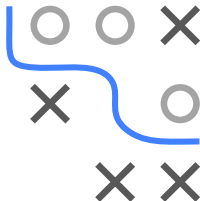
REMARKS

- It is also possible when the m -th task is multiclass classification. That is, $f(\mathbf{x})_m \in \mathbb{R}^{g_m}$ is the probability predictions for g_m classes.
~> The techniques for multi-target learning are also applicable under this setting, notation becomes cumbersome.
- Target space can be inhomogeneous, e.g. $\mathcal{Y}_m = \{0, 1\}$ and $\mathcal{Y}_k = \mathbb{R}$.
~> A mixture of multi-label classification and multivariate regression.



INDUCTIVE VS. TRANSDUCTIVE LEARNING

- In previous problems,
 - 1 predictions need to be generated for novel instances,
 - 2 targets are known beforehand and observed during training.
- These problems are *inductive* w.r.t. instances and *transductive* w.r.t. targets.



- Side information is important for generalizing to novel targets.
 - a novel target molecule in the drug design,
 - a novel tag in the document annotation,

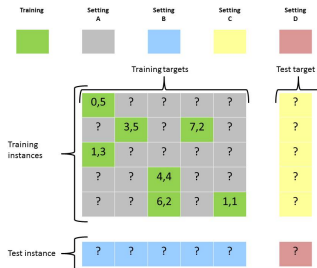
$g(\cdot, \cdot) : \text{target similarity}$

	Mol1	Mol2	Mol3	Mol4	Mol5	Mol6	Mol7
01101	1,3	0,2	1,4	1,7	3,5	1,3	?
00111	2	1,7	1,5	7,5	8,2	7,6	?
01110	0,2	0	0,3	0,4	1,2	2,2	?
10001	3,1	1,1	1,3	1,1	1,7	5,2	?
01011	4,7	2,1	2,5	1,5	2,3	8,5	?
11110	?	?	?	?	?	?	?

Waegeman et al. (2019), Multi-target prediction: A unifying view on problems and methods ([URL](#)).

SUBDIVISION OF DIFFERENT LEARNING SETTINGS

- Setting A — transductive w.r.t. targets and instances. Goal: predict missing values of score matrix (*matrix completion*).
- Setting B — transductive w.r.t. targets and inductive w.r.t. instances (*classical supervised learning*).
- Setting C — inductive w.r.t. targets and transductive w.r.t. instances.
~> Some targets are unobserved during training but may appear at prediction time.
- Setting D — inductive w.r.t. both targets and instances (*zero-shot learning*).



Waegeman et al. (2019), Multi-target prediction: A unifying view on problems and methods ([URL](#)).

